AWS VPC Peering & Private Network Verification

Project Technical Workbook: Private Inter-VPC Connectivity & Bastion Routing

STUDENT NAME	BATCH IDENTIFIER	TARGET AWS REGION	PROJECT STATUS
Ganesh Bharat Sherkhane	AWS-ARCH-2026	ap-south-1 (Mumbai)	• COMPLETED & VALIDATED

01. EXECUTIVE PROJECT SCOPE & ARCHITECTURE

The primary objective of this project is to architect an isolated dual-VPC topology within AWS, demonstrating end-to-end non-transitive inter-VPC private communication without crossing the public internet. Two distinct subnets in `ewb-vpc-a` and `ewb-vpc-b` exchange traffic across an AWS VPC Peering connection (`pcx-0e031f990cb0015e5`), with administrative ingress managed through a public Bastion host.



02. RESOURCE INVENTORY & ADDRESSING ALLOCATION

LOGICAL NAME	RESOURCE TYPE	VPC / SUBNET ASSOCIATION	IPV4 CIDR / ASSIGNED IP	AWS GENERATED ID
ewb-vpc-a	Virtual Private Cloud	Region: ap-south-1	10.10.0.0/16	`vpc-0c847e4e6f35eee76`
ewb-vpc-b	Virtual Private Cloud	Region: ap-south-1	10.20.0.0/16	`vpc-0e7899e2cec4b8011`
ewb-a-public	Public Subnet	ewb-vpc-a	10.10.1.0/24	`subnet-063290154c7185ac5`
ewb-b-private	Private Subnet	ewb-vpc-b	10.20.1.0/24	`subnet-06343e1c8f6861962`
ewb-peering-a-to-b	VPC Peering Connection	Requester: VPC A / Accepter: VPC B	Non-overlapping /16 CIDRs	`pcx-0e031f990cb0015e5`
ewb-instance-a	EC2 Compute (t3.micro)	ewb-vpc-a / ewb-a-public	10.10.1.229 (Pub: 65.2.151.248)	`i-04fd16cc1f69c6266`
ewb-instance-b	EC2 Compute (t3.micro)	ewb-vpc-b / ewb-b-private	10.20.1.81 (Private Only)	`i-0820b4b09bf79e619`

03. EXPLICIT ROUTING TABLE CONFIGURATIONS

VPC Peering connections do not automatically route traffic. Each route table associated with subnets on either side of the link must carry an explicit route entry pointing to the peering identifier (`pcx-0e031f990cb0015e5`). Below are the active route configurations applied to establish bidirectional reachability.

ROUTE TABLE NAME & ID	ASSOCIATED SUBNET	DESTINATION CIDR	TARGET	OPERATIONAL INTENT
ewb-rt-a-public `rtb-065c41b7fa63354e8`	`ewb-a-public` (10.10.1.0/24)	10.10.0.0/16	`local`	Local intra-VPC A communication
		0.0.0.0/0	`igw-08077cb346bc27c4e`	Egress/Ingress for Bastion administration
		10.20.0.0/16	`pcx-0e031f990cb0015e5`	Routes cross-VPC packets destined for VPC B into Peering Link
ewb-rt-b-private `rtb-05e29e0253bb56b6e`	`ewb-b-private` (10.20.1.0/24)	10.20.0.0/16	`local`	Local intra-VPC B communication
		10.10.0.0/16	`pcx-0e031f990cb0015e5`	Symmetric return path back to VPC A workloads

04. SECURITY GROUP RULES & LEAST PRIVILEGE CONTROL

Security groups operate as stateful firewalls at the ENI hypervisor layer. All rules are constrained to enforce least privilege access. Workload B permits no unrestricted public access (`0.0.0.0/0`).

SECURITY GROUP	TARGET INSTANCE	PROTOCOL / PORT	PERMITTED SOURCE	SECURITY JUSTIFICATION
ewb-sg-a `sg-0eb305c45e69e7f7b`	ewb-instance-a	SSH / TCP 22	0.0.0.0/0	Enables EC2 Instance Connect browser-based console terminal
ewb-sg-a `sg-0eb305c45e69e7f7b`	ewb-instance-a	All ICMP - IPv4	10.20.0.0/16	Permits incoming ICMP reply / verification packets from VPC B
ewb-sg-b `sg-0ff52243bb09c3209`	ewb-instance-b	SSH / TCP 22	10.10.0.0/16	Permits administrative access only from internal VPC A Bastion
ewb-sg-b `sg-0ff52243bb09c3209`	ewb-instance-b	All ICMP - IPv4	10.10.0.0/16	Permits ICMP Echo requests initiated from Instance A across Peering

> **Architectural Note: Statefulness of AWS Security Groups**
> AWS Security Groups are stateful. When an ICMP Echo Request or TCP connection is permitted inbound, the corresponding outbound return traffic is automatically allowed regardless of outbound rule restrictions. Default Network ACLs (stateless allow-all) were maintained on both subnets to avoid rule masking.

05. VERIFICATION TESTING & DIAGNOSTIC PROOFS

Testing was executed directly from the connected terminal on `ewb-instance-a` (Bastion node at `10.10.1.229`) targeting the private interface of `ewb-instance-b` at `10.20.1.81`.

TEST 01: PRE-PEERING BASELINE ATTEMPT (SIMULATED CONTROL)
Node: ewb-instance-a (10.10.1.229) → Target: 10.20.1.81 (Prior to Route Table peering entry)

```
[ec2-user@ip-10-10-1-229 ~]$ ping -c 4 -W 2 10.20.1.81 PING 10.20.1.81 (10.20.1.81) 56(84) bytes of data. --- 10.20.1.81 ping statistics --- 4 packets transmitted, 0 received, 100% packet loss, time 3072ms # RESULT: Baseline failure verified. Packets dropped at route table boundary with no target route.
```

TEST 02: POST-PEERING FORWARD PATH TRAVERSAL (ACTUAL VALIDATION LOG)
Node: ewb-instance-a (10.10.1.229) → Target: 10.20.1.81 (Via pcx-0e031f990cb0015e5)

```
[ec2-user@ip-10-10-1-229 ~]$ ping -c 4 10.20.1.81 PING 10.20.1.81 (10.20.1.81) 56(84) bytes of data. 64 bytes from 10.20.1.81: icmp_seq=1 ttl=127 time=0.654 ms 64 bytes from 10.20.1.81: icmp_seq=2 ttl=127 time=0.249 ms 64 bytes from 10.20.1.81: icmp_seq=3 ttl=127 time=0.180 ms 64 bytes from 10.20.1.81: icmp_seq=4 ttl=127 time=0.244 ms --- 10.20.1.81 ping statistics --- 4 packets transmitted, 4 received, 0% packet loss, time 3112ms rtt min/avg/max/mdev = 0.180/0.331/0.654/0.188 ms
```

06. CONTROLLED FAILURE & SECURITY VALIDATION

To confirm that security group boundaries independently enforce layer-4 perimeter security over the layer-3 peering route, a controlled security validation was performed:

```
# Phase 1: Inbound ICMP rule temporarily revoked on ewb-sg-b [ec2-user@ip-10-10-1-229 ~]$ ping -c 4 -W 2 10.20.1.81 --- 10.20.1.81 ping statistics --- 4 packets transmitted, 0 received, 100% packet loss, time 3065ms # Phase 2: Inbound ICMP Echo rule restored on ewb-sg-b (Source: 10.10.0.0/16) [ec2-user@ip-10-10-1-229 ~]$ ping -c 2 10.20.1.81 64 bytes from 10.20.1.81: icmp_seq=1 ttl=127 time=0.252 ms 64 bytes from 10.20.1.81: icmp_seq=2 ttl=127 time=0.211 ms --- 10.20.1.81 ping statistics --- 2 packets transmitted, 2 received, 0% packet loss, time 1002ms
```

Validation Confirmation: The controlled fault isolated the root cause to security group ingress inspection, proving that underlying Layer-3 VPC peering and routing table associations remained fully operational during the block.

07. TECHNICAL DEFENSE & ARCHITECTURE VIVA

Comprehensive technical defense answers required for formal AWS Architect Associate evaluation:

1. Why is the explicit return route required in VPC B's route table?
AWS VPC routing is destination-based and strictly symmetric. While the route table in VPC A directs outbound packets destined for `10.20.0.0/16` into the peering connection, VPC B has no inherent awareness of how to return reply packets unless `ewb-rt-b-private` explicitly directs traffic destined for `10.10.0.0/16` back to `pcx-0e031f990cb0015e5`. Without this return route, packets black-hole at the destination subnet boundary.

2. Why does App-B see the Bastion's private IP as its source address?
When connecting from a Bastion host (or via SSH proxying), the TCP connection to the target instance originates directly from the Bastion's private elastic network interface (`10.10.1.229`). Because the network packet is assembled locally within AWS, the target instance receives the packet with the Bastion's internal IP in the IPv4 source header field, never exposing the client's public internet IP.

3. Why does an "Active" peering connection status not guarantee ping success?
An "Active" peering state only confirms that the bidirectional logical handshake between the two VPC control planes has been accepted. Data packets cannot traverse the link until three decoupled layers align: (a) Route tables on both sides must contain explicit routes targeting the peering ID, (b) Subnet Network ACLs must permit bidirectional transit, and (c) Destination Security Groups must explicitly allow ICMP traffic from the sender's IP.

4. Can VPC B access the public internet through VPC A's Internet Gateway?
No. AWS VPC peering is strictly non-transitive and does not support edge-to-edge routing. An Internet Gateway attached to VPC A cannot be used by a peered VPC B as an egress path. Providing internet access to VPC B without a direct gateway would require an intermediate NAT proxy, AWS Network Firewall, or AWS Transit Gateway with appliance routing.

08. TEARDOWN VERIFICATION & COST CONTROL

EXECUTION ORDER	RESOURCE CLASSIFICATION	RESOURCE IDENTIFIER	TEARDOWN STATUS
Step 1 (Compute)	EC2 Virtual Instances	`ewb-instance-a`, `ewb-instance-b`	TERMINATED (0 active instances)
Step 2 (Storage)	EBS Root Volumes & Key Pairs	`vol-*` (gp3), `EWB-LAB`	DELETED / CLEANED UP
Step 3 (Interconnect)	VPC Peering Connection	`ewb-peering-a-to-b` (`pcx-0e031f990cb0015e5`)	DELETED (Routes Purged)
Step 4 (Routing & SGs)	Security Groups & Route Tables	`ewb-sg-a`, `ewb-sg-b`, `ewb-rt-*`	DISSOCIATED & DELETED
Step 5 (Infrastructure)	Subnets, IGW & VPCs	`ewb-igw-a`, `ewb-vpc-a`, `ewb-vpc-b`	DETACHED & DELETED

> **Zero Chargeable Footprint Confirmed:** AWS Elastic IP console confirmed *"No Elastic IP addresses found in this Region"*, 0 active NAT Gateways, and 0 running EC2 instances in region `ap-south-1`, satisfying all lab completion criteria.